

Growth and Cultivation of Arabidopsis Plants in Soil

Overview

Cultivate Arabidopsis seedlings on soil pots with a standard protocol.

A. Materials

- a. 75% Ethanol
- b. ½ MS square plates
- c. Bleach
- d. Soil pots
- e. Sterile ddH₂O

B. Procedure

Phase 1: Seed sterilization and germination

1. Sterilization & sowing:
 - a. Sterilize Arabidopsis seeds with 75% ethanol for 2 min. Then, add 95% ethanol and allow the seeds to air-dry under sterile conditions. Once dry, sow the seeds on square plates containing ½ MS medium.
2. Arrangement & Growth:
 - a. Arrange the plates into cold treatment at 4°C in the dark for 3 days, then transfer them into white light and culture vertically for 7 days

Phase 2: Soil & Pot preparation

3. Sanitization (optional):
 - a. Using 1:100 diluted bleach for 10 minutes, presoak the pots and water trays to sanitize.
 - b. Rinse thoroughly with clean water multiple times until all bleach odor is completely gone.
 - i. To avoid strong chemical odors, it is recommended to be performed within a hood.
4. Soil Preparation:
 - a. Mix peat soil and autoclave it. Fill each of the 3-inch pots with soil, lightly pressing down until 80% full, and thoroughly water the surface, ensuring the entire soil body is fully moistened before starting seedling transplantation.

Phase 3: Transplantation

5. Seedling transfer
 - a. Using a finger, make a small hole (about one knuckle deep) in the center of the soil

- i. When planting more than 1 seed, space the needed amount of holes evenly across the soil surface.
- b. Select 1 uniform, healthy, 7-day-old seedling and gently lift it with tweezers
- c. Place the root against the sidewall of the hole
- d. Using a wash bottle, rinse soil over the roots to ensure that they are naturally covered
 - i. *AVOID* touching the root system directly with tweezers or fingers
 - ii. During this process use tweezers to support the seedling from beneath the cotyledons
 - 1. *Never clamp the seedling directly*

Phase 4: Acclimatization & Post-Transplant Care

- 6. Humidity management
 - a. Place the pots onto a water tray and add water until the bottom of the pots are submerged in around 0.5 cm deep water.
 - b. Cover with a transparent lid, maintaining humidity for 3 days
 - i. Day 1: Keep the cover fully sealed
 - ii. Day 2: Open the cover slightly to lower humidity
 - iii. Day 3: To ensure good circulation and fungal growth, remove the cover completely
- 7. Watering
 - a. Adjust the watering frequency based on growth chamber humidity.
 - i. Soil surface should undergo dry-wet cycles
 - 1. Keep the surface dry for no more than 1 day to prevent dehydration
- 8. Fertilization:
 - a. Dilute 花寶2號 (Hyponex No. 2 Fertilizer) to 1:1000 and fertilize by alternating between water and fertilizer with each watering cycle.
 - i. Both should be applied via subirrigation (tray bottom). Avoid direct watering on the soil surface unless the soil is too dry to absorb water from the bottom

Phase 5: Experimental Treatment & Harvesting

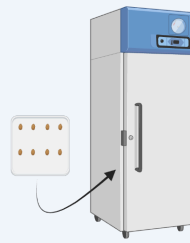
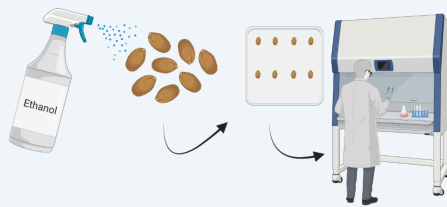
- 9. Experimental Treatment:
 - a. At approximately 3 weeks post-transplantation, the protectant/ stress treatments can be applied.
 - i. the timeframe can be adjusted between 2 to 4 weeks based on the plant growth
 - ii. Treatments should begin no later than 4 weeks post-transplant to avoid bolting and growth arrest
 - b. Take photos regularly (recommended daily) to track progress to identify optimal starting point
- 10. Support for Propagation

- a. Cover the sharp end of a bamboo stake with 200 μ L tip before inserting to prevent direct soil contact and mold growth
 - b. Once the inflorescence reaches 20 cm, insert the bamboo stake into the soil
 - c. Secure the inflorescence to the stake using orchid clips, preventing lodging
 - i. Make sure not to damage the stems with any clips
11. Seed harvesting → only for plants without any stress treatments
- a. Siliques will begin to dry about 1.5 months post-transplant; reduce watering frequency at this stage
 - b. Once 80% of all the siliques on the plant are dry, stop watering the plant completely to allow full drying before seed collection

Note: Plants not used for the experiments can be kept for seed propagation

Phase 1:

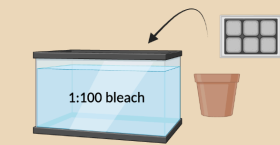
1) Using ethanol, sterilize the Arabidopsis seeds with ethanol and under sterile conditions, sow them in ½ MS square plates



2) Arrange the plates into cold treatment at 4°C in the dark for 3 days, then transfer them into white light and culture vertically for 7 days



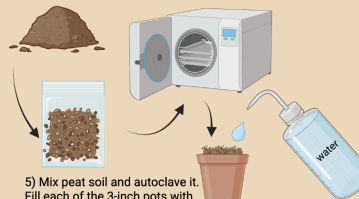
Phase 2:



3) Using 1:100 diluted bleach for 10 minutes, presoak the pots and water trays to sanitize.

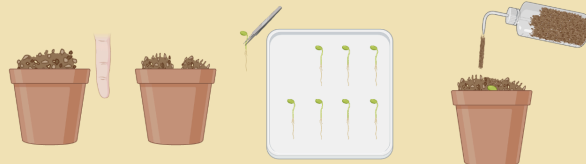


4) Rinse thoroughly with clean water multiple times until all bleach odor is completely gone.



5) Mix peat soil and autoclave it. Fill each of the 3-inch pots with soil, lightly pressing down until 80% full & fully moisten it

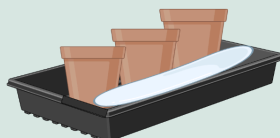
Phase 3.



5a, b, c. Use finger to make a small hole (one knuckle deep) and select one uniform 7-day-old seedling and place root against sidewall

5d. rinse soil over roots

Phase 4.



6a. Place pots onto water tray and add water until 0.5 cm deep



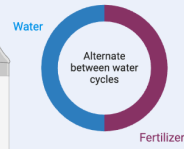
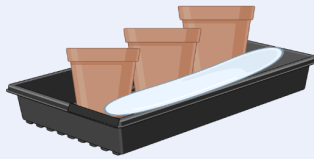
6b. add plastic lid and follow cycle:



Phase 4.



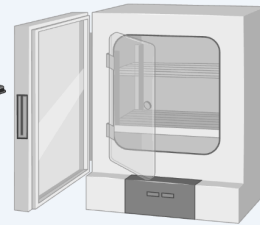
8a. Dilute
Hyponex
No. 2
Fertilizer
by 1:1000



8b. Fertilize the plants by alternating between water and fertilizer with each water cycle

Phase 5.

Three weeks post-transplantation



9. protectant and stress treatments may be applied